OBJECTIVES

Objective 1

Develop a cloud-based railway reservation system that intelligently manages ticket bookings, cancellations, and waiting-list operations using Reinforcement Learning techniques.

Objective 2

Reduce passenger waiting time by implementing an intelligent queue optimization mechanism that dynamically reallocates cancelled seats based on the learned policy of the Reinforcement Learning agent.

Objective 3

Improve seat utilization by minimizing vacant seats resulting from ticket cancellations through automated and optimized seat reallocation.

Objective 4

Deploy the proposed framework using AWS cloud services such as Amazon EC2, Amazon RDS, Amazon S3, AWS Lambda, and Amazon SNS to provide a scalable, secure, and reliable reservation management system.

Objective 5

Provide real-time reservation status updates and booking notifications to passengers through an automated notification service, improving user experience and communication.

Objective 6

Evaluate the performance of the proposed system using metrics such as average passenger waiting time, seat utilization rate, queue processing time, and successful reservation allocation efficiency.

Novelty Summary

The proposed Reinforcement Learning-Based Intelligent Queue Optimization Framework for Railway Reservation Systems enhances the traditional railway reservation process by integrating Reinforcement Learning (Q-Learning) with a cloud-based architecture. Unlike existing systems that mainly rely on static First-Come-First-Serve (FCFS) rules and predefined waiting-list mechanisms, the proposed framework introduces intelligent queue optimization, automated seat allocation, and scalable cloud deployment.

1. New Feature

The system introduces an **AI-assisted waiting-list management module** that dynamically reallocates cancelled seats using a Reinforcement Learning agent. This enables smarter reservation decisions compared to conventional static allocation methods.

2. Better Algorithm

A **Q-Learning algorithm** is used to learn from booking and cancellation patterns. The learned policy helps optimize seat allocation, reduce passenger waiting time, and improve overall reservation efficiency.

3. Better Architecture

The application follows a **cloud-based modular architecture** consisting of a React frontend, Node.js/Express backend, relational database, and Reinforcement Learning module. This modular design improves maintainability, scalability, and future extensibility.

4. Better AWS Integration

The proposed framework utilizes **Amazon EC2** for application hosting, **Amazon RDS** for reservation data, **Amazon S3** for storing datasets and logs, **AWS Lambda** for automated seat reallocation, **Amazon SNS** for reservation notifications, and **Amazon CloudWatch** for application monitoring.

5. Better Security

User authentication, secure database storage, and role-based access ensure that passenger information and reservation records are protected. Cloud-based storage also improves data reliability and availability.

6. Better Automation

The system automatically processes seat cancellations, updates the waiting list, and notifies passengers without manual intervention. This reduces processing time and improves operational efficiency.

7. Better Accuracy and Scalability

By learning from historical reservation data, the RL-based system provides more effective queue management than traditional rule-based methods. The cloud architecture also allows the system to handle increasing numbers of users and reservation requests with minimal performance degradation.